Cybersecurity Lab 1 · ELTE · Hamdan Al-Jabri · October 2026

Post-quantum key exchange: Part 1 summary

How fast are the new quantum-safe methods, and do they work correctly? (experiments E1 and E4)

1 Which methods we compare, and why these four

Post-quantum methods are built on two kinds of hard maths problems: **lattices** and **error-correcting codes**. Each kind comes in a **structured** version (extra patterns make it fast and compact, but give attackers more to work with) and an **unstructured** version (slower and bigger, but considered safer). We took one method from each of the four boxes:

	Lattice problems	**Code problems**
Structured (fast, compact)	**ML-KEM** the official US standard (FIPS 203, 2024)	**HQC** chosen by NIST in 2025 as the backup
Unstructured (cautious)	**FrodoKEM** ISO standard (2026), recommended in Germany	**Classic McEliece** ISO standard (2026), oldest unbroken design (1978)

This shows two things at once: lattices vs. codes, and **what the extra caution costs**. Each method is tested at three strength levels (NIST levels 1, 3 and 5, as strong as AES-128, -192 and -256), giving 12 versions. They are compared with **X25519** (today's method) and the **hybrid X25519MLKEM768** (both together, used by browsers today).

2 What is a KEM?

All four methods are **KEMs** (key-encapsulation mechanisms). A KEM has three steps (Figure 1):

1. **KeyGen**: the client makes a key pair and sends the public part.
2. **Encaps**: the server picks a random secret and locks it with that public key. It sends the locked box (the "ciphertext").
3. **Decaps**: the client opens the box with its private key.

Now both sides hold the same 32-byte secret. Think of the client sending an open padlock that only it can unlock.

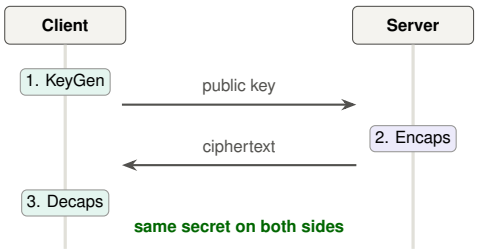


Figure 1. One key exchange with a KEM.

How X25519 fits. X25519 is not a KEM; both sides mix their own secret with the other's public value. It still has the same three-step shape: client key pair = KeyGen; server key pair + mixing = Encaps; client mixing = Decaps. So we measure it in exactly these roles (**X25519-ECDH**, as TLS uses it). We also measured the official "KEM-wrapped" version (**X25519-DHKEM**, RFC 9180). It adds one extra hashing step and costs 10–14% more, so plain ECDH is the fair baseline.

The hybrid sends both public parts together (1,184 + 32 = 1,216 bytes) and combines both secrets. It stays safe if either part is broken.

3 How we measured

- **Two computers, same software:** an **Apple M4** MacBook Air (ARM) and an **Intel i7-1185G7** laptop (x86). Both run Linux, the M4 in Docker and the Intel in WSL2, with the same compiler and libraries (liboqs 0.16.0, OpenSSL 3.5.8). Only the processor differs.
- **Three versions of the code: portable** (plain C that runs anywhere), **optimised** (uses special CPU instructions that process many numbers at once: AVX2 on Intel, NEON on ARM) and **OpenSSL** (the library real websites use, for X25519, ML-KEM and the hybrid). Optimised code exists for ML-KEM on both computers, and for FrodoKEM and McEliece only on Intel. HQC has none.
- **Timing:** every step is run 1,000 times and each run is timed. We report the **median** (the middle value, so a rare slow run does not distort it) and the **spread** of the middle half of the runs.
- **Fair conditions:** a 200 ms warm-up before timing (the M4 needs about 90 ms to reach full speed), a 30-second pause between methods (the fanless Air must cool down), and a measurement of the processor's actual speed. Results can then also be given in **CPU cycles**, the unit used in published papers.
- **McEliece key generation is timed 100 times, not 1,000.** One run takes 40–500 ms, so 1,000 runs would add hours. Its time also varies a lot: it picks a random secret and tries to build the public key, which works only about 29% of the time. Otherwise it starts again. The number of retries is luck, so one run can take six times as long as the next. 20 samples gave an unstable median; 100 give a stable one.

4 E4: is the code correct?

Yes: all 69 checks pass on both computers. The fast optimised code gives exactly the same results as the plain code, so the speed comparison is fair.

What is checked	Applies to	M4: passed	Intel: passed
1. Answers match known correct results	all 12 KEMs, both versions	24/24 (24 cases)	24/24 (24 cases)
2. NIST official FIPS 203 test vectors	ML-KEM only	6/6 (360 cases)	6/6 (360 cases)
3. Random exchanges give the same secret	all 12 KEMs, both versions	24/24 (918 cases)	24/24 (918 cases)
4. Keys from one version work with the other	all 12 KEMs	12/12 (24 cases)	12/12 (24 cases)
5. liboqs and OpenSSL work together	ML-KEM only	3/3 (6 cases)	3/3 (6 cases)
Total		**69/69 (1,332 cases)**	**69/69 (1,332 cases)**

Table 1. Correctness checks (E4). "All 12 KEMs" = 4 methods × 3 strength levels; "both versions" = portable and optimised code. **Each row is explained in Appendix A.** Not covered: NIST's input-validation tests, which liboqs does not expose.

5 E1: results

One whole key exchange (KeyGen + Encaps + Decaps, best code version). "vs X25519" is the time compared with X25519 on the same computer: below 1.0× means faster than X25519.

Method	Level	Bytes sent	Time (M4)	vs X25519	Time (Intel)	vs X25519
X25519 (today)	–	64	62.5 µs	1.0×	115 µs	1.0×
Hybrid X25519+ML-KEM-768	–	2,336	100 µs	1.6×	191 µs	1.7×
ML-KEM-512	1	1,568	17.6 µs	0.28×	20.8 µs	0.18×
ML-KEM-768	3	2,272	28.0 µs	0.45×	33.2 µs	0.29×
ML-KEM-1024	5	3,136	42.4 µs	0.68×	46.4 µs	0.40×
HQC-1	1	6,674	3.41 ms	55×	7.35 ms	64×
HQC-3	3	13,492	10.4 ms	166×	21.6 ms	189×
HQC-5	5	21,658	23.7 ms	379×	53.6 ms	467×
Frodo-640	1	19,368	2.72 ms	43×	2.23 ms	19×
Frodo-976	3	31,424	6.03 ms	97×	4.61 ms	40×
Frodo-1344	5	43,216	11.1 ms	178×	7.58 ms	66×
McEliece-348864	1	261,216	52.6 ms	841×	38.7 ms	337×
McEliece-460896	3	524,316	166 ms	2,652×	113 ms	983×
McEliece-6688128	5	1,045,200	287 ms	4,585×	134 ms	1,166×

Table 2. Bytes sent and time for one key exchange. Level = NIST strength level (1, 3, 5 ≈ AES-128, -192, -256).

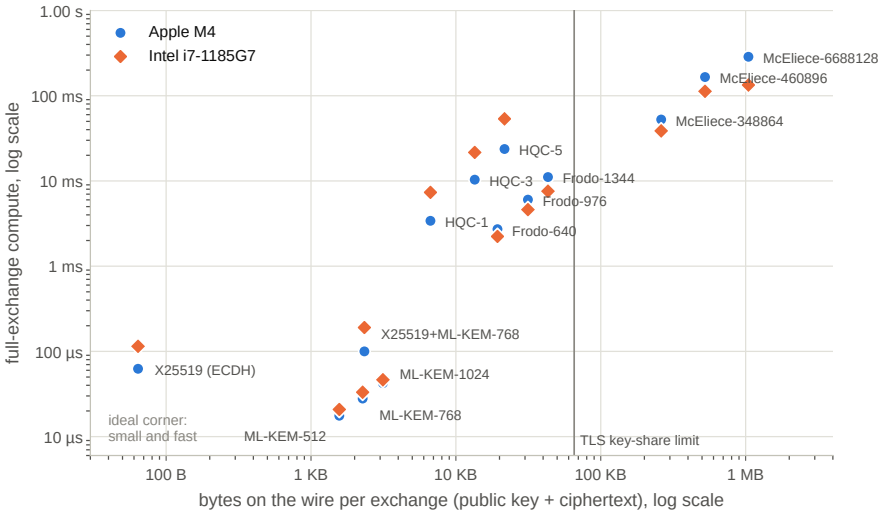


Figure 2. The same results as a picture: time vs. bytes sent. Bottom-left is best (fast and small). Methods right of the grey line do not fit in one TLS message.

6 What we found

1. **ML-KEM is faster than today's X25519** at every strength level: 2.2× (M4) and 3.5× (Intel) at level 3, and still faster at level 5. Its cost is **size**, not time: it sends 25–49× more bytes, about 1.5–3 KB. The reason: ML-KEM works on many small 12-bit numbers that a CPU handles in batches, while X25519 needs thousands of slow steps on 255-bit numbers.
2. **The hybrid costs 1.6–1.7× X25519**, almost exactly X25519 plus ML-KEM added together, so combining them costs nothing extra.
3. **The cautious methods are much more expensive, in both time and size.** At level 1: HQC 55–64× the time of X25519, FrodoKEM 19–43×, McEliece 337–841×. They send 4–14× more bytes than ML-KEM; McEliece's public

	Apple M4			Intel i7		
Method	KeyGen	Encaps	Decaps	KeyGen	Encaps	Decaps
X25519 (today)	13.2 μs	31.2 μs	18.0 μs	27.4 μs	57.4 μs	29.9 μs
Hybrid X25519+ML-KEM-768	30.1 μs	38.9 μs	31.0 μs	59.1 μs	73.7 μs	57.7 μs
ML-KEM-512	5.21 μs	5.67 μs	6.71 μs	5.85 μs	6.52 μs	8.46 μs
ML-KEM-768	8.50 μs	8.96 μs	10.5 μs	9.60 μs	10.3 μs	13.2 μs
ML-KEM-1024	12.8 μs	13.7 μs	15.9 μs	13.3 μs	14.6 μs	18.6 μs
HQC-1	561 μs	1.12 ms	1.72 ms	1.16 ms	2.36 ms	3.83 ms
HQC-3	1.69 ms	3.53 ms	5.14 ms	3.58 ms	7.17 ms	10.9 ms
HQC-5	3.72 ms	7.56 ms	12.4 ms	8.80 ms	18.1 ms	26.7 ms
Frodo-640	873 μs	927 μs	918 μs	681 μs	785 μs	767 μs
Frodo-976	1.91 ms	2.07 ms	2.05 ms	1.44 ms	1.60 ms	1.56 ms
Frodo-1344	3.57 ms	3.79 ms	3.75 ms	2.58 ms	2.59 ms	2.41 ms
McEliece-348864	38.6 ms	32.6 μs	13.9 ms	38.6 ms	13.0 μs	36.5 μs
McEliece-460896	141 ms	73.8 μs	24.7 ms	113 ms	29.3 μs	73.1 μs
McEliece-6688128	239 ms	121 μs	47.7 ms	134 ms	39.4 μs	68.0 μs

Table 3. Each step separately: median time per call, best code version. For McEliece, KeyGen is the slow step; for ML-KEM and HQC, Decaps is slowest.

	Intel i7 (AVX2)			Apple M4 (NEON)			Memory
Method	KeyGen	Encaps	Decaps	KeyGen	Encaps	Decaps	(KiB)
ML-KEM-512	3.2×	3.2×	3.0×	1.6×	1.5×	1.5×	10
ML-KEM-768	3.2×	3.0×	2.8×	1.7×	1.4×	1.5×	15
ML-KEM-1024	3.5×	3.1×	2.8×	1.7×	1.4×	1.4×	21
HQC-1	1.1×	1.0×	1.0×	1.0×	1.0×	1.0×	62
HQC-3	1.0×	1.0×	1.0×	1.0×	1.0×	1.0×	127
HQC-5	1.0×	1.0×	1.0×	1.0×	1.0×	0.9×	204
Frodo-640	2.9×	2.8×	2.8×	1.0×	1.0×	1.0×	75
Frodo-976	3.1×	3.0×	3.0×	1.0×	1.0×	1.0×	116
Frodo-1344	3.1×	3.3×	3.6×	1.0×	1.0×	1.0×	160
McEliece-348864	1.8×	3.6×	511×	1.3×	1.0×	1.0×	412
McEliece-460896	2.1×	3.1×	574×	1.0×	0.9×	1.0×	864
McEliece-6688128	3.7×	3.9×	1,405×	1.7×	1.0×	1.0×	1,529

Table 4. Optimised code and memory. Speed-up = time of the plain version / time of the optimised version (1.0× = no optimised code exists). Memory = peak working memory (stack) of the most demanding step.

key (0.26–1 MB) is too big for TLS. Moving from level 1 to 5 multiplies the cost by about 2.3 for ML-KEM and by 3.4–7 for the others.

4. **The code version matters as much as the method.** Optimised Intel code makes ML-KEM and FrodoKEM about 3× faster and McEliece's Decaps 510–1,400× faster. On the M4, only ML-KEM has optimised code (1.4–1.7× faster). So the M4 is 2.2× faster than the Intel laptop on plain code but slower for FrodoKEM and McEliece. liboqs has no optimised HQC; HQC's designers report code 33–49× faster than what we measured.
5. **Memory:** ML-KEM needs only 10–21 KB of working memory; HQC and FrodoKEM 60–200 KB; McEliece up to 1.5 MB, too much for small devices.
6. **Processor speed changes during the test:** the Intel CPU slows from 3.9 to 3.2 GHz while running optimised code; the M4 stayed between 4.0 and 4.44 GHz.
7. **Our numbers agree with published ones** for HQC, McEliece Encaps/Decaps and all key sizes. ML-KEM and FrodoKEM are faster than the 2021 figures (newer code). **One open question:** McEliece-348864 key generation takes 2.6× the published time.

Limits: both computers run Linux inside a virtual machine; one processor of each kind; the Intel time conversion varied 2.4% between sessions (CPU-cycle results are not affected). **Next (Part 2):** real TLS connections under network delay and loss (E2), and tests for secret-dependent timing leaks (E3). All details and tables are in the full Part 1 report; code and data at `github.com/hsh1994-ux/pq-kem-study`. Prepared with help from Claude (Anthropic); the design and interpretation are the author's.

Appendix A The five correctness checks (Table 1)

"**All 12 KEMs**" means the 4 methods, each at 3 strength levels: ML-KEM-512/768/1024, HQC-1/3/5, FrodoKEM-640/976/1344 and Classic McEliece-348864/460896/6688128. "**Both versions**" means the check ran once with the portable code and once with the optimised code.

1. **Answers match known correct results** (known-answer tests). *What:* liboqs ships a fingerprint (hash) of the exact output each method must produce when started from a fixed, known "random" seed. We run our build from the same seed, hash its output and compare. *Shows:* our build computes exactly the official algorithm, bit for bit. *Count:* 12 KEMs $\times$ 2 versions = 24 checks, one comparison each.
2. **NIST official FIPS 203 test vectors** (NIST ACVP). *What:* NIST publishes test cases for ML-KEM: "given these inputs, the right key, ciphertext or shared secret is exactly this". We feed in every input and compare our output with NIST's answer, for KeyGen, Encaps and Decaps. *Shows:* our ML-KEM matches the standard itself, judged by the standard's author. *Count:* 60 cases per level $\times$ 3 levels $\times$ 2 versions = 6 checks, 360 cases. ML-KEM only, because the other three methods have no NIST standard yet.
3. **Random exchanges give the same secret** (round trips). *What:* with fresh random inputs, run KeyGen $\rightarrow$ Encaps $\rightarrow$ Decaps and check that both sides end with the same secret. *Shows:* the full key exchange works with real random inputs, not only with fixed test cases. *Count:* 50 exchanges per KEM and version (McEliece: 3, because its key generation is slow) = 24 checks, 918 cases.
4. **Keys from one version work with the other** (cross-build). *What:* make a key with the portable code, encapsulate with the optimised code, decapsulate with the portable code; then the same in the other direction. *Shows:* the fast and the plain versions implement exactly the same algorithm. A version that was only consistent with itself would pass check 3 but fail here. This is what makes the speed comparison fair. *Count:* 12 KEMs $\times$ 2 directions = 12 checks, 24 cases.
5. **liboqs and OpenSSL work together** (interoperability). *What:* a key made by liboqs is encapsulated by OpenSSL and decapsulated by liboqs; then the reverse. *Shows:* two independently written implementations of the ML-KEM standard agree, so a client using one library can talk to a server using the other, as on the real internet. *Count:* 3 levels $\times$ 2 directions = 3 checks, 6 cases. ML-KEM only, because OpenSSL implements only ML-KEM among the four methods.

Total: 69 checks and 1,332 cases on each computer, all passed on both. Checks 1–2 prove that the code matches the reference and the standard; check 3 that it works with real random inputs; checks 4–5 that different implementations agree with each other.